

How these sheets work

One sheet per competency, 9 sheets for this video. Each gives you two prompts beside one open box. You build one prompt in the box. The other is there if you want more practice before the exam.

Before you start

1. Pick two pens you can tell apart. Pass one is the course notes, before the video. Pass two is a second color, after the concept video, into the same box.
2. Stop pass one while part of the box is still empty. That empty part is what the video fills in.
3. Read your prompt first, then divide the box to fit it. Some prompts want one labeled drawing, some a grid, some three panels. The division is yours.
4. Words go inside the drawing, as labels and inside little boxes with arrows. A sentence running the width of the box does not count.

Why two prompts

A and B are never the same question twice. One usually asks for a labeled drawing and the other for a grid or a sequence, so whichever way you think, there is a way in. Neither is the easy option.

About the paper

You do not have to print this. Rule one box per competency onto your own paper, or set it up on a tablet, and write the prompt letter you chose at the top. Hand ruled counts exactly the same.

This is anatomy

Every prompt asks what a structure is, where it sits, and what it is next to. Function belongs to physiology and is not on these sheets.

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8

WEEK 5 · AXIAL SKELETON, SKULL

Cranial and facial bones

COMP

You should be able to: Distinguish the eight cranial bones from the fourteen facial bones and name the region each bone forms.

PICK ONE AND DO THAT ONE

- ☐ A

LABELED SKULL, LATERAL VIEW
Draw a skull from the side. Label every cranial bone you can see and every facial bone you can see, and use two different marks, one for cranial and one for facial, so the two groups read at a glance. Write the count for each group.

Then: Draw a small anterior view beside it and add the facial bones that only show from the front.
- ☐ B

TWO-COLUMN SORT
Draw two columns headed cranial (8) and facial (14). List every bone, marking which are paired and which are single. Beside each bone, one phrase for the region it forms: the forehead, the cheek, the roof of the mouth, and so on.

Then: Circle the single unpaired bones in each column. There are fewer than you think.

PASS 1 COLOR _____

PASS 2 COLOR _____

ALSO A LAB SKILL

DRAW IT

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9

COMP

WEEK 5 · AXIAL SKELETON, SKULL

Significant skull bone markings

You should be able to: Identify the key markings on each cranial and facial bone, including features of the sphenoid, ethmoid, temporal, occipital, maxillae, and mandible.

PICK ONE AND DO THAT ONE

- ☐ **A BONE BY BONE PANELS**
Divide the box into six panels: sphenoid, ethmoid, temporal, occipital, maxilla, mandible. In each, draw the bone on its own, out of the skull, and label its key markings. The sphenoid and ethmoid panels need the view from above; the mandible needs the view from the side.

Then: In each panel, star the one marking you would use to identify that bone if it were handed to you loose.
- ☐ **B MARKING HUB MAP**
Draw six hubs, one per bone, spread across the box. From each hub, spoke out to its markings with a tiny drawing of the marking at the end of each spoke: the sella turcica as a saddle, the mastoid process as a bump, the crista galli as a fin, and so on.

Then: Draw one line between two hubs for a marking that two bones share or meet at, and name it.

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10

WEEK 5 · AXIAL SKELETON, SKULL

Sutures and fontanelles

COMP

You should be able to: Locate the four major sutures and the pterion by the bones they join and name the four fontanelles and where each lies.

PICK ONE AND DO THAT ONE

☐ **A TWO SKULLS, ADULT AND NEWBORN**

Draw an adult skull from above and from the side. Draw and label the coronal, sagittal, lambdoid, and squamous sutures, and mark the pterion where the four bones meet. Then draw a newborn skull in the same two views and label the anterior, posterior, sphenoidal, and mastoid fontanelles.

Then: Draw an arrow from each fontanelle to the adult suture or junction it becomes.

☐ **B JOINING TABLE**

Draw a table with the four sutures and the pterion down the side. Across the top: the bones it joins, where on the skull it runs, and a tiny sketch of its line. Below that, a second table for the four fontanelles: where each lies, which sutures meet there, and roughly when it closes.

Then: Circle the fontanelle a clinician checks on a baby and the junction a clinician worries about after a blow to the side of the head.

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Dr. Sharilyn Rennie

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11

COMP

WEEK 5 · AXIAL SKELETON, SKULL

Orbit, nasal cavity, palate, and sinuses

You should be able to: Identify the bones forming the orbit, nasal septum, and hard palate and locate the four paranasal sinuses by the bone each occupies.

PICK ONE AND DO THAT ONE

☐ **A THREE LABELED DRAWINGS**

Draw the orbit from the front and label every bone in its walls. Draw the nasal septum from the side and label the two bones and the cartilage that form it. Draw the hard palate from below and label the two bones that form it, front and back.

Then: On the orbit drawing, shade the four sinuses in the bones around it and name each.

☐ **B BONE-TO-SPACE GRID**

Draw a grid with the bones down the side: frontal, ethmoid, sphenoid, maxilla, palatine, lacrimal, zygomatic, vomer, nasal. Across the top: orbit, nasal septum, hard palate, paranasal sinus. Tick every cell where that bone helps form that space.

Then: Beside the grid, draw a face outline and mark where each of the four sinuses sits.

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COMP

WEEK 5 · AXIAL SKELETON, SKULL

Major skull foramina and contents

You should be able to: Name the major skull foramina by the bone each pierces and state the nerve or vessel that passes through each.

PICK ONE AND DO THAT ONE

☐ **A LABELED CRANIAL FLOOR**

Draw the floor of the cranial cavity from above, the three cranial fossae as steps. Draw each major foramen as a hole in the right fossa, label it, and write beside it the bone it pierces and what passes through.

Then: Draw the foramen magnum last, larger than the rest, and label what passes through it.

☐ **B THREE-COLUMN TABLE**

Draw a table with the foramina down the side: foramen magnum, optic canal, superior orbital fissure, foramen ovale, cribriform foramina, internal acoustic meatus, carotid canal, jugular foramen, mental foramen. Across the top: the bone, the fossa or region, and what passes through.

Then: Group the rows by bone with a bracket, so the sphenoid rows sit together.

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COMP

WEEK 5 · AXIAL SKELETON, SPINE

Vertebral column regions and curvatures

You should be able to: Name the five regions of the vertebral column with the vertebra count of each and identify the four curvatures as primary or secondary.

PICK ONE AND DO THAT ONE

- ☐ **A** LABELED SPINE, LATERAL VIEW
- Draw the whole vertebral column from the side. Bracket the five regions and write the count in each. Draw the four curvatures clearly, and mark each one primary or secondary with a different mark.
- Then: Beside the column, draw a tiny curled newborn spine to show which curves are there from the start.
- ☐ **B** REGION AND CURVE TABLE
- Draw a table with the five regions down the side and across the top: vertebra count, fused or separate, the curvature in that region, and whether that curve is primary or secondary. Draw the curve's shape in the cell instead of describing it.
- Then: Add a row at the bottom for the total count of vertebrae in an adult and in a child, and show where the difference comes from.

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14

COMP

WEEK 5 · AXIAL SKELETON, SPINE

Typical vertebra and intervertebral disc

You should be able to: Label the parts of a typical vertebra and the anulus fibrosus and nucleus pulposus of an intervertebral disc.

PICK ONE AND DO THAT ONE

☐ **A TWO LABELED VIEWS**

Draw a typical vertebra from above and label the body, vertebral arch, pedicles, laminae, spinous process, transverse processes, vertebral foramen, and the superior articular processes. Then draw two vertebrae from the side with the disc between them and label the anulus fibrosus and nucleus pulposus.

Then: On the side view, mark the intervertebral foramen and label what passes through it.

☐ **B PART HUB**

Draw a vertebra small in the middle of the box. Spoke out to every part and, at the end of each spoke, draw that part on its own and write what it is: a pedicle as a short bridge, a lamina as a plate, and so on. Add a second small hub for the disc with its two parts.

Then: Write beside the disc hub which of the two parts is the tough ring and which is the gel.

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15

COMP

WEEK 5 · AXIAL SKELETON, SPINE

Regional and specialized vertebrae

You should be able to: Distinguish cervical, thoracic, and lumbar vertebrae by an identifying feature and describe the atlas, axis, dens, sacrum, and coccyx.

PICK ONE AND DO THAT ONE

- ☐ A

IDENTIFICATION PANELS

Draw a cervical, a thoracic, and a lumbar vertebra from above, side by side, and circle on each the feature that gives it away. Below them draw the atlas and the axis from above, label the dens, and draw the sacrum and coccyx from the front.

Then: Beside the atlas, write which bone sits on it and which bone the dens belongs to.
- ☐ B

GIVEAWAY TABLE

Draw a table with cervical, thoracic, lumbar, atlas, axis, sacrum, coccyx down the side. Across the top: the one feature that identifies it, the shape of its body, what its spinous process looks like, and a tiny sketch. Fill the sketch column with drawings, not words.

Then: Put a box around the two rows that have no body at all, and write why the atlas is missing one.

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16

WEEK 5 · AXIAL SKELETON, SPINE

Sternum and ribs

COMP

You should be able to: Identify the parts of the sternum and rib markings and classify the twelve rib pairs as true, false, or floating by anterior attachment.

PICK ONE AND DO THAT ONE

☐ **A LABELED THORACIC CAGE**

Draw the thoracic cage from the front. Label the manubrium, body, and xiphoid process of the sternum, and the sternal angle. Number the twelve ribs and use three marks to show true, false, and floating. Draw the costal cartilages so the anterior attachment of each rib is visible.

Then: Draw one rib on its own below and label the head, neck, tubercle, and shaft.

☐ **B ATTACHMENT SORT**

Draw three columns headed true, false, floating. In each, list the rib numbers and draw how that group reaches, or fails to reach, the sternum. Below the columns, draw the sternum on its own and label its three parts and the sternal angle.

Then: Mark the rib that attaches at the sternal angle, and write why that landmark is useful for counting ribs.

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